

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Smart agriculture production optimization engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.N o	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses only historical weather data and manual input	Prediction accuracy may decrease for sudden weather changes	high
2	No real-time weather API integration	Users must manually enter weather values, making predictions less convenient	high
3	Limited scalability and cloud deployment	The system may not efficiently support a large number of simultaneous users	medium

Scalability Plan:

S.N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	supports a limited number of users on a local Flask server	Deploy on IBM Cloud/AWS with load balancing and auto-scaling to support thousands of users

2	Data Storage	Stores dataset locally (CSV/SQLite)	Use a cloud database such as IBM Db2, MySQL, or PostgreSQL for secure and scalable storage
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3	Performance	Prediction is done on a single server with a pre-trained model	Optimize the model, use caching, and deploy with Gunicorn/Nginx or Docker for faster response
4	Security	Basic Flask application with minimal security	Implement HTTPS, user authentication, API security, input validation, and encrypted data storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	
Phase 3	
Phase 4	